

Project Design Phase

Problem – Solution Fit Template

Date	30 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? I.e. working parents of 0-5 y.o. kids	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking	Explore AS, differentiate
Focus on J&P, tap into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides.	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? I.e. customers have to do it because of the change in regulations.	7. BEHAVIOUR What does your customer do to address the problem and get the job done? I.e. directly related: find the right solar panel installer, calculate usage and benefits; Indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)	Focus on J&P, tap into BE, understand RC
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? I.e. lost, insecure → confident, in control - use it in your communication strategy & design.		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	

Define customer segments, fit into customer limitations

1. CUSTOMER SEGMENT(S) Add Small and medium farmers. Agricultural extension officers. Agricultural researchers.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES Add No expert access. Low tech knowledge. Low digital literacy. Severe time limits.	5. AVAILABLE SOLUTIONS PLUSES & MINUSES Add Traditional farming knowledge . Local agricultural experts. Government advisory services. Manual soil analysis.
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Focus on problem, tap into behavior, understand root cause

2. PROBLEMS / PAINS + ITS FREQUENCY Add Hard to pick right crop. Low farm output.	9. ROOT / CAUSE OF PROBLEM Add Limited access to smart Systems	7. BEHAVIOR + ITS INTENSITY Add Asks nearby farmers. Checks online/soil logs.
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No data-backed choices.

No automated crop tips.

Identify strong triggers & emotions

3. TRIGGERS

Add

Low crop yield.

Crop picking doubts.

Weather Changes

4. EMOTIONS

Add

Before: Anxious & lost.

After: Sure & satisfied

10. YOUR SOLUTION

Add

OptiCrop: Flask-based ML web application✕

Uses soil nutrient and environmental input

Trained ML prediction model.

Provides crop recommendations.

8. CHANNELS OF BEHAVIOR

ONLINE

Add

Agricultural websites, YouTube.

mobile/web applications.

OFFLINE

Add

Agricultural offices,.

Local experts.